Xuanjun Chen

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Summary

Xuanjun is a Ph.D. student at National Taiwan University, advised by Prof. Hung-yi Lee and Prof. Jyh-Shing Roger Jang. His research focuses on **Speech/Audio Processing** and **Large Language Models**, specifically audio understanding, generation, deepfakes, and agentic LLMs, with support from NVIDIA Academic Grant and IEEE SPS Scholarship. He has published over 20 peer-reviewed papers, **15 as first/co-first author**, across top ML/NLP (ICLR, COLM, ACM TIST, ACL Findings) and speech (TASLP, ICASSP, INTERSPEECH, etc.) venues, earning a **best paper award** at ROCLING 2025 and nominations at IEEE ASRU 2025 and ISCSLP 2026.

Education

Ph.D. Candidate in Communication Engineering, **National Taiwan University**, GPA: 4.3/4.3 — Sep 2023 – Present
M.S. in Computer Science and Information Engineering (CSIE), **National Taiwan University**, GPA: 4.19/4.3 — Sep 2020 – Jan 2023
B.S. in CSIE, **National Taiwan University of Science and Technology**, GPA: 4.11/4.3 — Jun 2020

Research Experience

NVIDIA Academic Grant Program, Lead Student Researcher, Taipei Taiwan — Oct 2026 – Mar 2027 (expected)

- **Audio Generation.** Leading research on fine-grained text-to-audio model to enhance audio generation capabilities.

Shanda Group, Research Intern (Remote), Tokyo Japan — Apr 2026 – Jul 2026

- **Speech Generation.** Investigated whether LLM agents can automate reinforcement learning for Text-to-Speech models. [1]

National Taiwan University, Research Assistant (Ph.D. & M.S.), Taipei Taiwan — Oct 2020 – Present

- Published **15 first/co-first** peer-reviewed papers, with 5 more currently under review.
- **Speech and Audio Language Models.** Analyzed the post-training dynamics of large audio language models [2], authored a high-impact survey [7], and served as a core contributor to DeSTA2.5-Audio [3], Dynamic-SUPERB [4], and Codec-SUPERB [5] [6].
- **Retrieval and Agentic LLMs.** Pioneered a dual-loop agentic framework for optimizing benchmark generation [8]; improved retrieval via graph navigation [9] and delta planning [10], alongside a systematic RAG study for historical archives [11].
- **Audio Deepfakes.** Conducting research on speech anti-spoofing, building upon prior work on audio deepfake detection across various scenarios [12]–[21] and audio-visual learning [22]–[24], including a system ranked 3rd of 42 teams in ASVspoof 2021 (LA track) [25].

Research Highlights

Total citations **600+** *as of Oct 2026.*

- MetaBench-Harness: Unlocking End-to-End Optimization of Benchmark Harnesses — *ICLR '27 (Submitted)*
  TL;DR: *A dual-loop framework where a meta-harness refines the harnesses that evolve benchmarks hard enough for frontier LLMs.*
- Can LLM Agents Automate Reinforcement Learning for Text-to-Speech? — *ICASSP '27 (Submitted)*
  TL;DR: *An LLM agent reconstructs and improves an RL post-training recipe for zero-shot TTS, cutting failure cases by half.*
- What Does LALM Post-training Learn? Rethinking Audio Understanding and Reasoning Capabilities — *NAACL '27 (Submitted)*
  TL;DR: *Much of the post-training gain comes from fitting answer formats and textual cues, not from better audio understanding.*
- Only Ask What You Don't Know: Grounded Delta Planning for Efficient Multi-step RAG — *COLM 2026*
  TL;DR: *Plans only for missing information after a first retrieval; top multi-hop QA accuracy at 22–68% lower cost.*
- CodaRAG: Connecting the Dots with Associativity Inspired by Complementary Learning — *ACM TIST 2026*
  TL;DR: *Brain-inspired RAG that chains dispersed evidence via associative graph navigation; +7–10% recall on GraphRAG-Bench.*
- CodecFake+: Codec-Based Resynthesized Data as a Proxy for Detecting CodecFake Speech — *TASLP 2026*
  TL;DR: *Re-synthesized speech from 31 codecs plus a codec taxonomy as a training proxy for detecting 17 CoSG deepfake systems.*
- Towards Generalized Source Tracing for Codec-Based Deepfake Speech — *ASRU 2025* · Best Student Paper Nominee
  TL;DR: *SASTNet fuses Whisper semantics with Wav2vec2/AudioMAE acoustics to trace CodecFake speech back to its source codec.*
- DeSTA2.5-Audio: Toward General-Purpose Large Audio Language Model with Self-Generated Cross-Modal Alignment — *TASLP 2026* ‡
  TL;DR: *General-purpose LALM trained on 5M self-generated audio QA pairs; keeps the backbone LLM's language ability.*
- Dynamic-SUPERB Phase-2: A Collaboratively Expanding Benchmark for Spoken Language Models with 180 Tasks — *ICLR 2025* ‡
  TL;DR: *180 community-contributed tasks over speech, music, and audio; no current model handles all of them well.*
- Codec-SUPERB: An In-Depth Analysis of Sound Codec Models — *ACL Findings 2024* †
  TL;DR: *Benchmarks neural audio codecs on resynthesis quality and downstream speech, speaker, emotion, and sound tasks.*

First/co-first author unless marked † (co-second author) or ‡ (contributing author).

Selected Publications

* *equal contribution;* † *co-second author;* ‡ *contributing author*

Speech and Audio Language Models and Generation

[1] **X. Chen**, et al., "Can LLM Agents Automate Reinforcement Learning for Text-to-Speech?" *ICASSP 2027 (Submitted)*

[2] H.H. Chen*, **X. Chen***, et al., "What Does LALM Post-training Learn? Rethinking Audio Understanding and Reasoning Capabilities," *NAACL 2027 (Submitted)*

[3] K.H. Lu, Z. Chen, ... **X. Chen**‡, ... H. Lee, "DeSTA2.5-Audio: Toward General-Purpose Large Audio Language Model with Self-Generated Cross-Modal Alignment," *TASLP, 2026*

[4] C.Y. Huang, ... **X. Chen**‡, ... S. Watanabe, H. Lee, "Dynamic-SUPERB Phase-2: A Collaboratively Expanding Benchmark for Measuring the Capabilities of Spoken Language Models with 180 Tasks," *ICLR 2025*

[5] H. Wu*, ... **X. Chen**†, ... H. Lee, "Codec-SUPERB: An In-Depth Analysis of Sound Codec Models," *Findings of ACL 2024*

[6] H. Wu, **X. Chen**, et al., "Codec-SUPERB @ SLT 2024: A Lightweight Benchmark for Neural Codec Models," *SLT 2024*

[7] H. Wu, **X. Chen**, et al., "Towards Audio Language Modeling – An Overview," *Technical Report, Feb. 2024*

Retrieval and Agentic Large Language Models

[8] **X. Chen**, et al., "MetaBench-Harness: Unlocking End-to-End Optimization of Benchmark Harnesses," *ICLR '27 (Submitted)*

[9] C.Y. Li*, **X. Chen***, et al., "CodaRAG: Connecting the Dots with Associativity Inspired by Complementary Learning," *ACM TIST, 2026*

[10] W.C. Chou*, **X. Chen***, et al., "Only Ask What You Don't Know: Grounded Delta Planning for Efficient Multi-step RAG," *COLM 2026*

[11] C. Lin*, B.H. Feng*, **X. Chen***, et al., "A Preliminary Study of RAG for Taiwanese Historical Archives," *ROCLING 2025* · Best Paper Award

Audio Deepfakes: Neural Codec, Singing Voice, Adversarial Attack, Conventional TTS/VC, and Audio-Visual Scenarios

[12] **X. Chen**, et al., "Why Is Spectrogram ResNet Still Effective in Singing Voice Deepfake Detection?" *ICASSP '27 (Submitted)*

[13] Y.A. Hsieh*, **X. Chen***, et al., "On Bona Fide Audio Discrepancies in Singing Voice Deepfake Detection," *ICASSP '27 (Submitted)*

[14] **X. Chen***, et al., "CodecFake+: Codec-Based Resynthesized Data as a Proxy for Detecting CodecFake Speech," *TASLP, 2026*

[15] **X. Chen**, et al., "Mitigating Proxy-to-Wild Domain Gap in Deepfake Speech," *ISCSLP 2026* · Selected Outstanding Paper

[16] **X. Chen***, C.Y. Hu*, et al., "Joint Fullband-Subband Modeling for High-Resolution SingFake Detection," *INTERSPEECH 2026* · Long Oral

[17] **X. Chen**, et al., "How Does Instrumental Music Help SingFake Detection?" *ICASSP 2026*

[18] **X. Chen***, I.M. Lin*, et al., "Towards Generalized Source Tracing for Codec-Based Deepfake Speech," *ASRU 2025* · Best Student Paper Nominee

[19] **X. Chen**, et al., "Codec-Based Deepfake Source Tracing via Neural Audio Codec Taxonomy," *INTERSPEECH 2025*

[20] **X. Chen**, et al., "Singing Voice Graph Modeling for SingFake Detection," *INTERSPEECH 2024* · Oral

[21] **X. Chen**, et al., "Neural Codec-based Adversarial Sample Detection for Speaker Verification," *INTERSPEECH 2024*

[22] **X. Chen**, et al., "Localizing Audio-Visual Deepfakes via Hierarchical Boundary Modeling," *ISCSLP 2026*

[23] **X. Chen**, et al., "Multimodal Transformer Distillation for Audio-Visual Synchronization," *ICASSP 2024*

[24] **X. Chen**, et al., "Push-Pull: Characterizing the Adversarial Robustness for Audio-Visual Active Speaker Detection," *SLT 2022*

[25] Y.L. Liao*, **X. Chen***, et al., "Adversarial Speaker Distillation for Countermeasure Model on Automatic Speaker Verification," *SPSC 2022*

Selected Honors, Service & Skills

Honors

- **NVIDIA Academic Grant Recipient** GPU compute resources via competitive selection by NVIDIA — 2026
- **IEEE SPS Scholarship** One of dozens of recipients worldwide from the IEEE Signal Processing Society — 2026
- **Google Student Travel Grant** Sponsored by Google Asia Pacific to present at ICASSP 2024 — 2024
- **CTCI Foundation Sci. & Tech. Scholarship** Bursary Award 2024 & Research Scholarship 2025 — 2024–2025
- **Dean's List** Certificate of Achievement for the top 5% of CSIE students, Taiwan Tech — Fall '18, Fall '19, Spr. '20

Service

- **Co-Organizer,** Responsible Speech and Audio Generative AI, Special Session at IEEE ASRU 2025 — 2025
- **Technical Committee,** Codec Speech Proc. Universal PERformance Benchmark Challenge, IEEE SLT 2024 — 2024
- **Reviewer:** AAAI, ICLR, NeurIPS, ICML; ACL, EMNLP, NAACL, COLM; ICASSP, INTERSPEECH, ASRU, SLT, etc. — 2023–Now

Skills

- **Coding & Toolkit:** Python, LaTeX; PyTorch, PyTorch Lightning, Hugging Face Transformers